

MAINTENANCE WORK ORDER LOG

Petromax Baytown Refining Complex
January 1, 2025 -- June 30, 2025

Document Number: LOG-WO-2025-H1
Prepared by: Maintenance Planning Department
System of Record: IBM Maximo Asset Management v7.6.1.3

1. Work Order Summary -- Q1 2025 (January - March)

WO #	Tag	Date	Description	Priority	Status	Hours
5001	P-101	01/22	Vibration analysis - elevated DE brg	HIGH	CLOSED	4.0
5002	E-301	01/28	ECT tube inspection - 12 tubes plugged	HIGH	CLOSED	16.0
5003	V-201	02/03	UT thickness survey - 24 CML locations	MED	CLOSED	8.0
5004	C-401	02/10	Valve inspection - 2nd stg disc valve	LOW	CLOSED	6.0
5005	TK-501	02/18	API 653 external inspection	MED	CLOSED	12.0
5006	P-101	02/25	Mech seal leak investigation	HIGH	CLOSED	3.0
5007	P-102	03/05	Routine PM - oil sample & vib check	LOW	CLOSED	2.5
5008	E-301	03/12	Performance test - U-value calc	MED	CLOSED	4.0

WO-5001: P-101 Vibration Analysis

Date: January 22, 2025

Equipment: P-101 -- Centrifugal Feed Pump

Requestor: M. Okonkwo, Vibration Analyst

Findings: Route-based vibration survey identified elevated readings at P-101 drive-end bearing position. Overall velocity measured 0.42 in/s peak, exceeding the Alert level of 0.35 in/s per the plant vibration severity chart (based on ISO 10816-3). Frequency spectrum shows dominant peak at 1x running speed (59.3 Hz = 3,560 RPM), indicating possible unbalance or misalignment. Sideband activity observed at bearing defect frequencies suggesting early-stage inner race degradation.

Action: Recommended mechanical seal and coupling inspection at earliest opportunity. Work Order WO-5006 raised for follow-up investigation. Monitoring frequency increased from monthly to weekly.

WO-5002: E-301 Eddy Current Testing

Date: January 28, 2025

Equipment: E-301 -- Shell & Tube Heat Exchanger

Requestor: R. Vasquez, Sr. Inspection Engineer

Findings: Eddy Current Testing (ECT) performed on 450 tubes using Eddyfi Ectane 2 instrument. Results: 12 tubes identified with >40% wall loss -- all plugged with tapered brass plugs per ASME PCC-2. An additional 23 tubes showed 20-40% wall loss, flagged for re-inspection at next outage. Total plugged tube count now 31/450 (6.9%). Fouling evident on shell side based on pressure drop increase (4.2 to 11.7 psi). Calculated fouling factor: 0.0035 hr-ft²-F/BTU versus design value of 0.001.

Action: Chemical cleaning recommended during Q2 turnaround. Work Order WO-5012 to be raised for cleaning scope. Engineering evaluation req'd if plugged count approaches 45 tubes (10%).

WO-5006: P-101 Mechanical Seal Leak Investigation

Date: February 25, 2025

Equipment: P-101 -- Centrifugal Feed Pump

Requestor: Operations (CRO shift report -- visible drip from seal gland)

Findings: Visual inspection confirmed active leakage from the outboard seal face of the John Crane Type 4620 dual mechanical seal. Leak rate quantified at approximately 12 drops/minute by timed drip test. API 682 acceptance criterion is 3 drops/minute for hydrocarbon service. Barrier fluid (Shell Thermia B) reservoir level dropping approximately 0.5 liters/day. Seal faces inspected via borescope -- visible scoring and heat checking on the outboard face. Probable root cause: operating at reduced suction pressure during the January cold snap caused brief cavitation events that damaged the seal faces.

Action: Seal replacement scheduled for April 2025 turnaround window. Follow SOP-MNT-001 Rev. 4 for the seal replacement procedure. In the interim, increase barrier fluid monitoring frequency to daily and maintain reservoir topped up. Pump operation approved to continue with restrictions (maximum 90% rated flow to minimize seal face loading).

2. Work Order Summary -- Q2 2025 (April - June)

WO #	Tag	Date	Description	Priority	Status	Hours
5010	P-101	04/07	Mech seal replacement per SOP-MNT-001	CRIT	CLOSED	24.0
5011	P-101	04/08	Laser alignment post-seal replacement	CRIT	CLOSED	6.0
5012	E-301	04/14	Shell-side chemical cleaning	HIGH	CLOSED	18.0
5013	V-201	04/20	Level instrument recalibration	LOW	CLOSED	2.0
5014	TK-501	05/05	Secondary seal partial replacement	MED	OPEN	Est. 40
5015	P-101	05/12	30-day post-repair vib survey	MED	CLOSED	3.0
5016	C-401	06/01	2nd stg discharge valve replacement	MED	PLANNED	Est. 8
5017	E-301	06/15	Post-cleaning performance test	MED	CLOSED	4.0

WO-5010: P-101 Mechanical Seal Replacement

Date: April 7, 2025

Equipment: P-101 -- Centrifugal Feed Pump

Procedure Reference: SOP-MNT-001 Rev. 4

Lead Technician: A. Ramirez, Maintenance Tech Level III

Work Performed:

- Pump isolated per LOTO-P101-001 (3 energy isolation points)
- Casing drained and nitrogen purged; atmosphere verified safe
- Removed John Crane Type 4620 dual mechanical seal (S/N: JC-46200-29847)
- Seal face inspection: outboard face exhibited severe heat checking and a radial crack approximately 0.3" long; inboard face showed moderate wear grooves (depth ~0.001")
- Shaft sleeve inspected: OD measured 3.4985" (nominal 3.500") -- within tolerance
- Seal chamber bore measured 4.8760" (nominal 4.875") -- within tolerance
- Installed new John Crane Type 4620 seal (S/N: JC-46200-31052)
- Gland plate torqued per specification (Superbolt jackbolts, 45 ft-lbs, star pattern, 3 passes)

- Barrier fluid system charged with Shell Thermia B, pressurized to 190 psig (25 psi above seal chamber)
- Coupling element (Rexnord Omega E40) replaced with new

Total labor: 24 hours (3 technicians x 8 hours)

Parts consumed: 1x JC Type 4620 seal kit, 1x Rexnord E40 coupling element, 5 gal Shell Thermia B

WO-5011: P-101 Laser Alignment

Date: April 8, 2025

Equipment: P-101 -- Centrifugal Feed Pump

Alignment Equipment: Pruftechnik ROTALIGN Ultra iS (S/N: RT-20-1847)

Technician: B. Thompson, Alignment Specialist

Pre-alignment (as-found):

Vertical offset: +0.012" (motor high)

Horizontal offset: -0.006" (motor left)

Vertical angularity: +0.0018"/inch

Horizontal angularity: -0.0009"/inch

Post-alignment (final):

Vertical offset: +0.006" (motor high, includes 0.005" thermal growth compensation)

Horizontal offset: +0.001" (within tolerance)

Vertical angularity: +0.0003"/inch (within tolerance)

Horizontal angularity: -0.0002"/inch (within tolerance)

Shim corrections: Added 0.008" to outboard motor feet, removed 0.004" from inboard motor feet. Motor slid 0.007" to the right. Total shim pack at each foot: 3 shims (within 4-shim maximum).

Result: PASS -- all values within tolerance per SOP-MNT-001 Section 7.

WO-5015: P-101 Post-Repair Vibration Survey (30-Day)

Date: May 12, 2025

Equipment: P-101 -- Centrifugal Feed Pump

Analyst: M. Okonkwo, Vibration Analyst CAT III

Results:

Drive-end bearing: 0.15 in/s peak velocity (1x) -- EXCELLENT

Non-drive-end bearing: 0.12 in/s peak velocity (1x) -- EXCELLENT

No bearing defect frequencies detected

Seal area vibration: within normal limits

Comparison to pre-repair: Drive-end bearing velocity reduced from 0.42 in/s to 0.15 in/s, a 64% improvement. New vibration baseline established and uploaded to AMS Machinery Manager.

Assessment: Mechanical seal replacement and precision alignment have fully resolved the elevated vibration condition. P-101 returned to normal monitoring frequency (monthly).